25901582 **MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

MGU - UGP (HONOURS) REGULAR EXAMINATION, March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEZGY200, MG4DSEZIM200 -

GENERAL TOXICOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

VALUATION SCHEME

Part A
Multiple Choice Questions (Answer all questions).
Each question carries 1 mark

1. Nicotine is classified as a toxicant of: [K] / [CO1]
b) Plant origin

2. Xenobiotics with which structural property cross biological membranes most easily by passive diffusion?
b) High lipid solubility [U] / [CO3]

3. The compound most frequently causing allergic contact dermatitis in hair dyes is
c) PPD [U] / [CO4]

4. Global economic burden of occupational injuries and diseases is estimated at [K] / [CO5]
a) 3 trillion dollars

5. Statement 1: Lipophilic compounds like PCBs concentrate in the blubber of marine mammals.
Statement 2: Contaminant levels in marine mammals can be thousands of times higher than in the surrounding seawater. [U] / [CO4]
c) Both statements are true

6. Which scientist is called the Father of Modern Toxicology? [K] / [CO1]
b) Mathieu Orfila

7. Corpora allata is a gland in insects that produces [K] / [CO3]
b)juvenile hormone

8. Which of the following is listed as part of "Supportive Care" for CWA victims? [U] / [CO4]
c) Airway, Breathing and Circulation (ABCs), oxygen supplementation and IV access

9. Aflatoxin is mainly produced by ……. [U] / [CO2]
a)Aspergillus flavus

10. DDT is an example of …… [U] / [CO3]
b) Persistent organic pollutant

Part - B
Short Answer Questions. Answer any **10** questions
Each question carries **3** marks.

11. What are good laboratory practices in toxicological studies? How is it important? [K]/[CO1]

Good Laboratory Practices (GLP) serve as a standardized quality framework designed to guarantee that laboratory research is dependable, consistent, and honest. By following OECD and national guidelines, GLP mandates that every phase of an experiment—ranging from the ethical treatment of animals to the final reporting of results—is executed with rigorous documentation and uniformity. They major features of GLP are : Strategic Study Design: Ensuring the research logic is sound from the start. Resource Management: Utilizing skilled staff, verified techniques, and precisely tuned equipment. Data Integrity: Maintaining exhaustive and precise records of all activities. Adhering to these standards does more than just a lab's scientific reputation; it is a mandatory prerequisite for regulatory bodies when evaluating chemical safety or approving new drugs. Furthermore, GLP compliance facilitates the international exchange of data, preventing the need for redundant testing across different countries. (2m - practices; 1m -importance)

12. What is the role of Phase I reactions in xenobiotic metabolism? [U] / [CO3]

Phase I reactions, primarily catalyzed by cytochrome P450 enzymes, serve as the initial step in xenobiotic metabolism by oxidizing, reducing, or hydrolyzing foreign compounds. Their main role is to convert lipophilic xenobiotics into more polar, water-soluble metabolites by introducing or exposing functional groups (e.g., OH, NH2, SH, COOH)

Phase I reactions introduce or expose functional groups and make compounds more chemically reactive and slightly more polar.

13. Classify the major classes of Cyanobacterial toxins. [U] / [CO4]

The major classes of cyanobacterial toxins are:

1. Hepatotoxins (Liver Toxins)

These are the most common cyanotoxins found in freshwater, targeting the liver and causing serious hemorrhagic damage.

2. Neurotoxins (Nerve Toxins)

These toxins affect the nervous system, causing rapid death by respiratory arrest (often acting in minutes).

3. Dermatotoxins and Cytotoxins (Skin/Cell Toxins)

These toxins cause inflammation, skin rashes, and irritation, typically upon contact with benthic (bottom-dwelling) cyanobacterial mats. (Each - 1m with definition)

14. What are mechanical hazards? [U] / [CO5]

Mechanical hazards are physical dangers posed by moving machine parts, tools, or materials that can cause injuries like crushing, cutting, shearing, or entanglement. They occur when workers contact unguarded, rotating, or falling equipment, often involving machinery, vehicles, or high-pressure systems. Common examples include conveyors, gears, and sharp edges.(3/4 points)

15. List different proactive engineering measures giving examples [K] / [CO5]

Proactive engineering measures are systematic, preventive actions taken to anticipate and mitigate hazards, equipment failures, or performance issues before they occur. These measures, often falling under the "Engineering Controls" tier of the hierarchy of hazard controls, aim to alter the physical environment or equipment to enhance safety, reliability, and efficiency. (4/5 points).

16. What is LD50? [K] / [CO1]

LD50 (Lethal Dose 50%) is a standard measure of acute toxicity, representing the amount of a substance—administered all at once—required to kill 50% of a test population (usually mice or rats). A lower LD50 indicates higher toxicity. It is typically expressed in milligrams per kilogram of body weight.

17. How xenobiotics redistributed in the body? [U] / [CO3]

Redistribution of xenobiotics refers to the movement of xenobiotics from one tissue to another after their initial distribution. This process occurs due to changes in blood flow, tissue affinity, and chemical nature of the xenobiotic. Initially, xenobiotics are delivered to highly perfused organs such as the brain, liver, lungs, and kidneys. Over time, xenobiotics may move to less perfused tissues such as adipose tissue, muscle, and bone, where they can be stored. Lipid-soluble xenobiotics show greater redistribution because of their affinity for fat tissues. Examples include DDT, PCBs, and dioxins, which accumulate in adipose tissue and are slowly released, causing prolonged and chronic toxic effects.(4/5 relevant points)

18. Explain Veterinary toxicology. [U] / [CO4]

Veterinary toxicology is the study of the harmful effects of chemical, biological, and physical agents on animals, focusing on identifying, diagnosing, and treating poisoning in domestic, wild, and aquatic species. It addresses the, diagnosis, treatment, and prevention of toxicosis, ensuring food safety and investigating environmental toxic risks to animals. (3/4 relevant poins)

19. Differentiate flammable and corrosive chemicals with suitable examples [U] / [CO5]

Flammable and corrosive chemicals represent two distinct categories of hazardous materials defined by how they interact with their environment and human tissue. **Flammable chemicals** pose a fire hazard by igniting easily, whereas **corrosive chemicals** cause damage by chemically destroying

living tissues or materials upon contact.

Flammable Chemicals: These substances catch fire easily when exposed to air, heat, or a spark. They are classified based on their "flash point"—the lowest temperature at which they produce enough vapor to ignite. **Examples:** Gasoline, Acetone, Ethanol (Alcohol), Benzene, Butane, Paint Thinners, Kerosene, Sodium, and Potassium.

Corrosive Chemicals: These substances cause immediate, severe damage upon contact with living tissue (eyes, skin) or destroy metals. They can cause burns, lesions, and permanent tissue damage. **Examples:** Hydrochloric Acid Sulfuric Acid , Sodium Hydroxide , Hydrofluoric Acid, Bromine, and Ammonia.(Definition of two - 2m, examples for both - 1m)

20. How do Agricultural pesticides cause mass mortality in birds? [U] / [CO4]

- **Ingestion of Treated Seeds & Insects:** Neonicotinoid pesticides, often coated on seeds, are highly toxic. Birds eating these seeds or insects that have consumed the pesticides are directly poisoned.
- **Acute Toxicity and Neurological Damage:** Pesticides act as neurotoxins, attacking the central nervous system. This causes immediate mortality or debilitating symptoms like loss of coordination and inability to fly.
- **Starvation and Anorexia:** Some pesticides, such as imidacloprid, induce severe anorexia in migrating birds, causing them to lose significant body weight and fat stores, making them unable to complete migration.
- **Food Source Depletion:** Pesticides kill the insects (caterpillars, beetles, flies) that birds rely on to feed their nestlings. The loss of prey, particularly for aerial insectivores (swallows, swifts), causes mass starvation of chicks.
- **Reproductive Failure:** Contamination affects eggshell quality, reduces breeding success, and impairs parental care, leading to reduced offspring survival.
- **Systemic Exposure:** Systemic insecticides are absorbed into all plant tissues, including nectar, pollen, and fruit, exposing birds that do not directly eat insects or seeds. (any 4 points)

21. How does lipid solubility influence absorption of Xenobiotics? [U] / [CO3]

Lipid solubility plays a crucial role in the absorption of xenobiotics because biological membranes are composed of a phospholipid bilayer. Xenobiotics that are highly lipid-soluble can easily dissolve in this membrane and pass through it by passive diffusion, without requiring energy or transport proteins. As a result, they are absorbed more rapidly and efficiently compared to water-soluble substances, which cannot easily penetrate the lipid-rich membrane and therefore show slower absorption.(relevant 3/4 points)

22. Explain the concept of Chemical Weapons. [U] / [CO4]

1. Highly toxic synthetic substances
2. Distinct from biological or nuclear weapons
3. Designed to incapacitate, injure or kill through their direct chemical effects on physiological processes.
4. These agents can disperse as gas, aerosol or liquid.
5. Route of entry - inhalation, Dermal and ingestion Eg: choking agents - chlorine and phosgene, sulfur mustard gas etc.(4/5 points)

Part C

Short Essay Questions Answer any **6** questions
Each question carries **5** marks

23. Summarize the A-B-C-D-E classification of adverse drug reactions. [U] / [CO4]

The two major classes of cyanobacterial toxins are Hepatotoxins (eg. Microcystins) and Neurotoxins (eg. Anatoxins).

1. Type A (Augmented)
   Most common. Predictable based on the drug's known mechanism of action .Dose dependent
2. Type B (Bizarre)
   Unpredictable, Specific to individuals. Not dose dependent

3. Type C (chronic)
   Associated with long term cumulative use of a drug
4. Type D (Delayed)
   Effects appear long after the use of the drug.
5. Type E (End of use/ Withdrawal)
   Reaction occur upon the abrupt discontinuation of a drug to which the body has become dependent (1m for each division with details)

24.Compare byssinosis and bagassosis. [A] / [CO5]

Byssinosis and bagassosis are occupational lung diseases caused by inhaling vegetable dusts. Byssinosis ("brown lung") arises from inhaling cotton, flax, or hemp dust, often causing "Monday fever" symptoms. Bagassosis is caused by breathing in moldy sugarcane waste (bagasse), leading to hypersensitivity pneumonitis. (2½m for each definition and differences)

25 **Explain sources and bioaccumulation of Mercury. Comment on its acute and chronic toxicity. [U] / [CO3]**

Mercury poisoning primarily results from consuming fish containing high levels of methylmercury, inhaling vapor from elemental mercury in broken devices (thermometers/switches), or using contaminated skin-lightening creams. Major environmental sources include coal combustion, small-scale gold mining, and industrial pollution.

Key sources of mercury exposure include:

- **Contaminated Seafood (Methylmercury):** High-mercury fish include shark, swordfish, king mackerel, tuna (especially albacore), orange roughy, marlin, and tilefish.
- **Industrial and Environmental Pollution:** Coal-burning power plants and industrial waste release mercury into the air, which settles into water and enters the food chain.
- **Broken Products (Elemental Mercury Vapor):** Thermometers, barometers, thermostats, and older fluorescent light bulbs contain elemental mercury.
- **Dental Amalgam Fillings:** Older "silver" fillings contain ~50% mercury by weight.
- **Gold Mining:** Small-scale, industrial, and artisanal gold mining often uses mercury to extract gold, leading to severe local exposure and environmental contamination.
- **Consumer Products:** Certain skin-lightening creams, soaps, and traditional medicines may contain mercury.

Those at higher risk include pregnant women, nursing mothers, young children, and people with high consumption of predatory fish. (2½m for sources and bioaccumulation, 2½m for toxicity)

26. How do food additives help improve the flavour, colour, texture, and overall appearance of food products? Cite suitable examples. [K] / [CO2]

Food additives enhance food by intensifying flavor, stabilizing textures, restoring or brightening colors, and boosting visual appeal. Key functions include adding savory taste, thickening consistency, creating uniform color, and preventing oxidation or drying. Common examples include preservatives, emulsifiers, sweeteners, and artificial colors. **Flavor Enhancers & Modifiers** MSG (Monosodium Glutamate): Used in savory products like soups, chips, and frozen meals to intensify the savory "umami" taste. ▪ Vanillin ▪ Esters ▪ Sweeteners (e.g., Aspartame, Sucralose, Stevia): Provide sweetness without the caloric density of sugar, often used in diet beverages. ▪ Acids (e.g., Malic Acid, Citric Acid): Add tartness and brighten flavors, balancing the overall taste. **Texture & Consistency Modifiers** ▪ Emulsifiers (e.g., Lecithin, Xanthan Gum Mono- and diglycerides): Prevent separation in products like salad dressings, peanut butter, and margarine, ensuring a smooth, combined consistency. ▪ **Stabilizers & Thickeners** (e.g., Gelatin, Starch, Guar Gum, Carrageenan, Pectin): Create a creamy texture in ice cream and prevent the formation of ice crystals. **Color & Appearance Improvers** ▪ Colorants (e.g., Tartrazine, Erythrosine, Sunset Yellow, Caramel): Restore color lost during processing or add vibrant hues to candy, jellies, and cereals. ▪ **Glazing Agents** (e.g., Waxes): Applied to fruits like apples to create a shiny, appealing surface. ▪ **Firming Agents** (e.g., Calcium Chloride): Used to keep processed fruits and vegetables, like canned tomatoes, firm. **Overall Appearance & Quality Preservation** ▪ Antioxidants (e.g., Ascorbic Acid/Vitamin C):

Prevent browning and rancidity in fruits, fats, and oils. ▪ **Anti-caking agents**: Ensure powdered products like salt and spices remain free-flowing and do not clump. ▪ **Preservatives (**e.g., Sodium Benzoate, BHA, BHT, Vitamin E): Extend shelf life, allowing food to retain a fresh, appealing look for a longer duration. (Different level of enhancers - 3m, examples - 2m)

27. What are essential heavy metals? Explain their role in physiological and biochemical processes. [U] / [CO3]

A sub group of heavy metals required by organisms in trace amounts for vital physiological and biochemical functions in the body. They serve as essential cofactors for numerous enzymes, playing crucial roles in catalysis, electron transport and the regulation of metabolic pathways. Their balanced concentration is essential for maintaining homeostasis, as both deficiency and excess can lead to metabolic disturbances and toxic effects. Eg – Iron, copper, zinc, manganese, cobalt etc
(Definition- 1mark, Functions – 3 marks, Any 2 examples – 1 mark)

28. Explain the standard drugs that are used to treat nerve agent poisoning. [U] / [CO4]

The standard treatment for nerve agent poisoning (e.g., Sarin, VX, Novichok) involves a three-pronged pharmacological approach designed to combat the "cholinergic crisis," which is caused by the inhibition of the enzyme acetylcholinesterase, leading to an overstimulation of acetylcholine receptors.

1. Atropine (Anticholinergic/Antimuscarinic)
2. Pralidoxime Chloride/2-PAM Cl (Oxime Reactivator)
3. Benzodiazepines (Anticonvulsants - e.g., Diazepam, Midazolam)

Patients known to have been exposed to nerve agents or those exhibiting symptoms of nerve agent exposure should be given the antidote medications atropine and pralidoxime. Atropine is a competitive antagonist of ACh at the neuromuscular junction and is the primary antidote in nerve agent poisoning.(Drugs names - 2m, explanation of each drug's mechanism, action, etc - 3m)

29. Define occupational cancer and write a short note on the major occupational cancers. [K] / [CO5]

Definition – The type of cancer which arises from long-term exposure to workplace carcinogens like smoke, various chemicals, solar radiation etc. (1 mark)
Major occupational cancers – skin cancer, lung cancer, bladder cancer, leukemia
Skin Cancer: Skin cancer involves uncontrolled growth of abnormal skin cells, typically due to ultraviolet (UV) radiation from the sun or artificial sources. The major types are basal cell carcinoma, squamous cell carcinoma and melanoma. Early signs include changes in existing moles, shiny bumps, rough scaly patches or lesions that itch or bleed. Preventive strategies include UV protection, protective clothing and minimizing exposure to known carcinogens.
Lung Cancer: Lung cancer develops when abnormal cells in the lungs multiply uncontrollably, forming tumors that destroy healthy tissue. While smoking is the leading cause, occupational exposure to carcinogens such as asbestos, radiation and second-hand smoke also increases risk. Early symptoms are often absent. Advanced disease presents with persistent cough, chest pain, hemoptysis, breathlessness, hoarseness, weight loss and fatigue. Lung cancers are categorized as: Non-small cell lung cancer (NSCLC) and Small cell lung cancer (SCLC)
Bladder Cancer: Bladder cancer begins in the urothelial lining of the bladder, where uncontrolled cell growth forms tumors. Early symptoms include hematuria, frequent urination, urgency and painful urination. Occupational exposure is a major contributor, particularly to aromatic amines found in the dye, rubber, chemical and paint industries Hairdressers, barbers and firefighters also face increased risk through chronic exposure to carcinogens. Types include - Urothelial carcinoma, Squamous cell carcinoma , Adenocarcinoma , Small cell carcinoma
Leukaemia: Leukaemia is a cancer of the blood and bone marrow involving uncontrolled growth of abnormal white blood cells. These cells fail to function properly, accumulate in the bone marrow and suppress normal blood cell production, leading to anemia, infection and bleeding tendencies.
Risk factors include ionizing radiation, benzene exposure and certain chemotherapeutic agents (alkylating agents such as cyclophosphamide and chlorambucil). Workers involved in chemical,

pharmaceutical, paint, plastic and oil industries may be increased risk due to benzene exposure. Occupational exposure to cytotoxic drugs dur preparation or administration also poses hazards to healthcare workers. Symptoms include easy bruising, bleeding gums, nosebleeds, frequent infection fatigue and pallor. Treatment varies by leukemia type, with chemotherapy remaining mainstay.
(Explanation on the types – 1 mark each for 2-3 relevant points)

30. Explain the contribution of industrial and environmental pollutants to xenobiotic exposure. [U] / [CO3]

Industrial and environmental pollutants are primary contributors to human xenobiotic exposure, which refers to the intake of foreign chemical substances—such as heavy metals, pesticides, plasticizers, and industrial by-products—that cannot be easily metabolized by the body. Rapid industrialization, urbanization, and agricultural intensification have led to widespread environmental contamination, resulting in chronic, low-dose exposure through air, water, soil, and food ingestion.

1. Major Sources of Environmental Pollutants

Industrial Emissions & Waste: Industries (manufacturing, mining, chemical, energy) release significant amounts of heavy metals (e.g., Pb, Hg, As, Cd), persistent organic pollutants (POPs), and volatile organic compounds (VOCs).

Agriculture & Pesticides: Extensive use of pesticides (e.g., glyphosate, organophosphates) and fertilizers introduces xenobiotics directly into the soil and water.

Plastic Pollution & Additives: Nano- and microplastics (NPs/MPs), arising from the fragmentation of larger plastic waste or industrial applications, are ubiquitous in water, air, and food.

Waste Management Practices: Inadequate disposal, including open burning of garbage and e-waste, releases toxic compounds into the atmosphere and soil.

2. Pathways of Human Exposure

Inhalation: Particulate matter (PM) and volatile compounds from vehicle exhaust and industrial emissions are inhaled, depositing toxins deep in the respiratory tract.

Ingestion: Contaminated groundwater, agricultural products (plants, vegetables), and seafood (contaminated with methylmercury or microplastics) are major dietary sources of xenobiotics.

Dermal Absorption: Direct skin contact with polluted soil, water, or chemical products in industrial settings.

3. Contribution, Mechanisms & Impacts

Bioaccumulation & Biomagnification: Pollutants like heavy metals and POPs persist in the environment and accumulate in organisms, resulting in high concentrations in top-level predators (including humans).

Particulate Matter (PM) as Carriers: Small particles, particularly PM2.5, act as "shuttles" that adsorb toxic substances (metals, PAHs) on their surface, facilitating their penetration into the bloodstream and distribution to distant organs.

Endocrine Disruption: Chemicals such as Bisphenol A (BPA) and phthalates, released from plastics, mimic or interfere with hormonal signaling, causing reproductive and metabolic disorders.

Epigenetic Alterations: Chronic exposure to pollutants causes long-term modifications in gene expression (e.g., DNA methylation), contributing to cancer, neurodegenerative diseases, and autoimmune conditions.

Microbiota Disturbance: Environmental contaminants, particularly heavy metals and dioxins, alter the gut microbiome composition, driving the selection of strains with degrading capacity and increasing antibiotic resistance.

The cumulative impact of these pollutants often leads to chronic diseases such as cancer, reproductive dysfunction, and immunotoxicity, highlighting the urgent need for enhanced environmental monitoring and pollution control. (Source, pathway and contribution- each explanation 1½m).